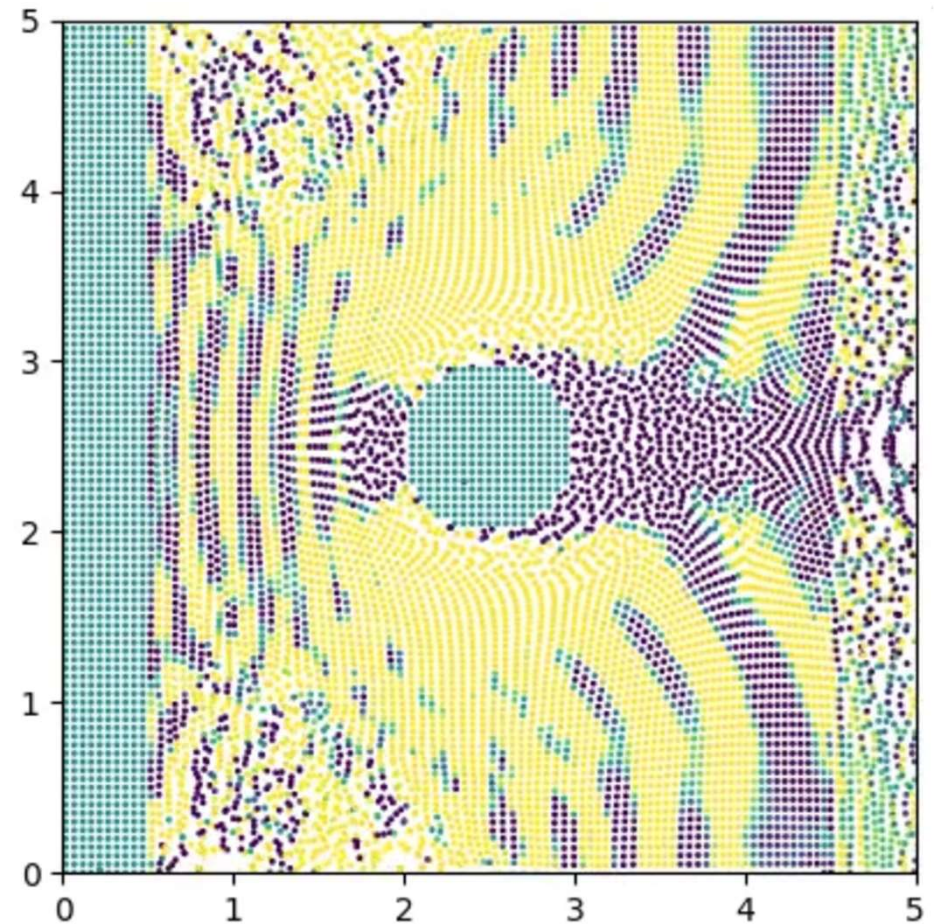


Calculating drag coefficients with **Smoothed Particle Hydrodynamics**

Mats Lenes (5141346)
Niels Swinkels (5182360)



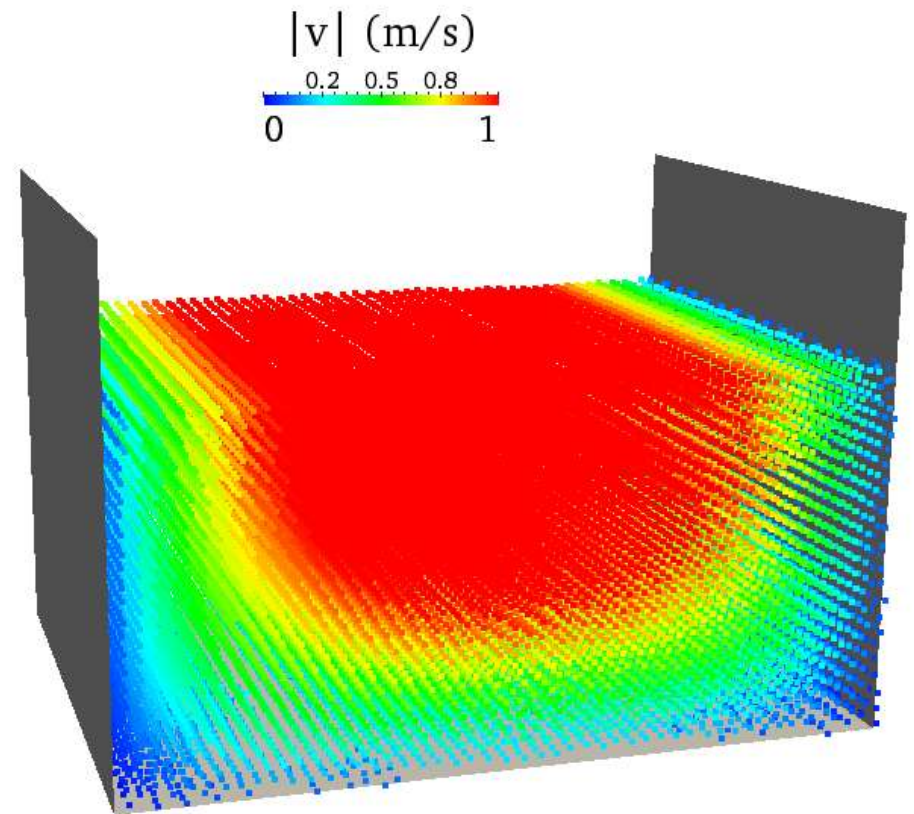
Smoothed Particle Hydrodynamics

Smoothing $\rightarrow$ particles behave like liquid

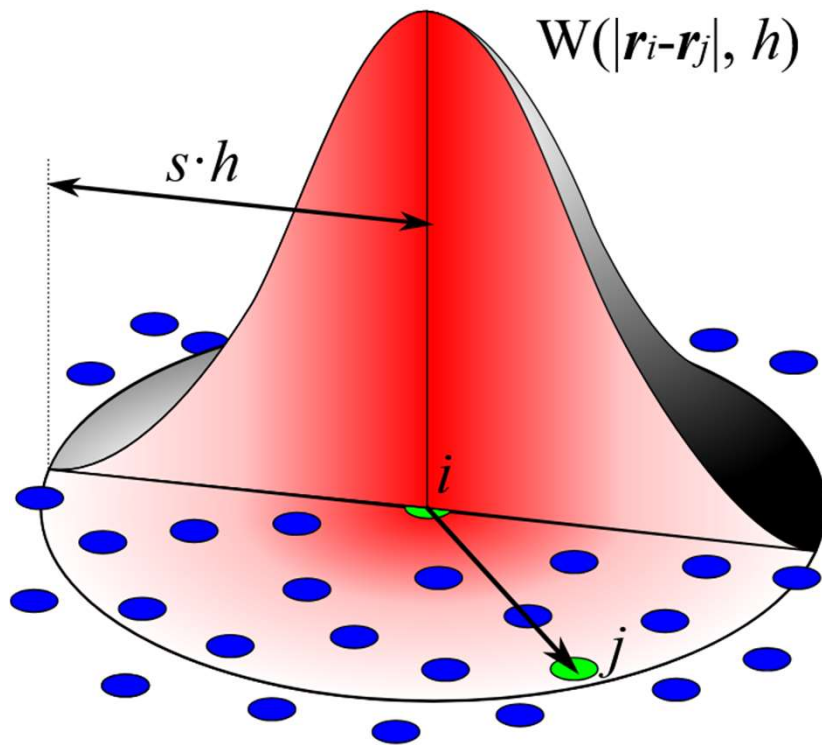
Computationally expensive $\rightarrow$ C++ instead of Python

Used to calculate drag coefficients, based on Korzani et. al. (2017)

$$F_D = \frac{1}{2} \rho |\mathbf{v}|^2 c_D A$$



Kernels & Quantity interpolation



Convolutional approximation:

$$Q(\mathbf{r}) \approx \int_V Q(\mathbf{r}') W(|\mathbf{r} - \mathbf{r}'|, h) dV$$

Riemann sum:

$$Q(\mathbf{r}) \approx \sum_i Q_i V_i W(|\mathbf{r} - \mathbf{r}'|, h)$$

Using $V_i = m_i / \rho(\mathbf{r})$:

$$Q(\mathbf{r}) \approx \sum_i Q_i \frac{m_i}{\rho(\mathbf{r})} W(|\mathbf{r} - \mathbf{r}'|, h)$$

Particles exert **forces** on each other

Pressure force:

$$\mathbf{F}_{i,\text{pres}} = -m_i \sum_j m_j \left(\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2} \right) \nabla W_{ij}$$
$$P_i = P_0 + c_s^2(\rho_i - \rho_0)$$


Viscous force:

$$\mathbf{F}_{i,\text{visc}} = -m_i \sum_j m_j \frac{\mu_i + \mu_j}{\rho_i \rho_j} \frac{\mathbf{v}_i - \mathbf{v}_j}{\mathbf{r}_i - \mathbf{r}_j} \frac{\partial W_{ij}}{\partial r_{ij}}$$
$$\mu_i = \nu \rho_i$$


Solid objects are made of **ghost particles**

Free particles move.

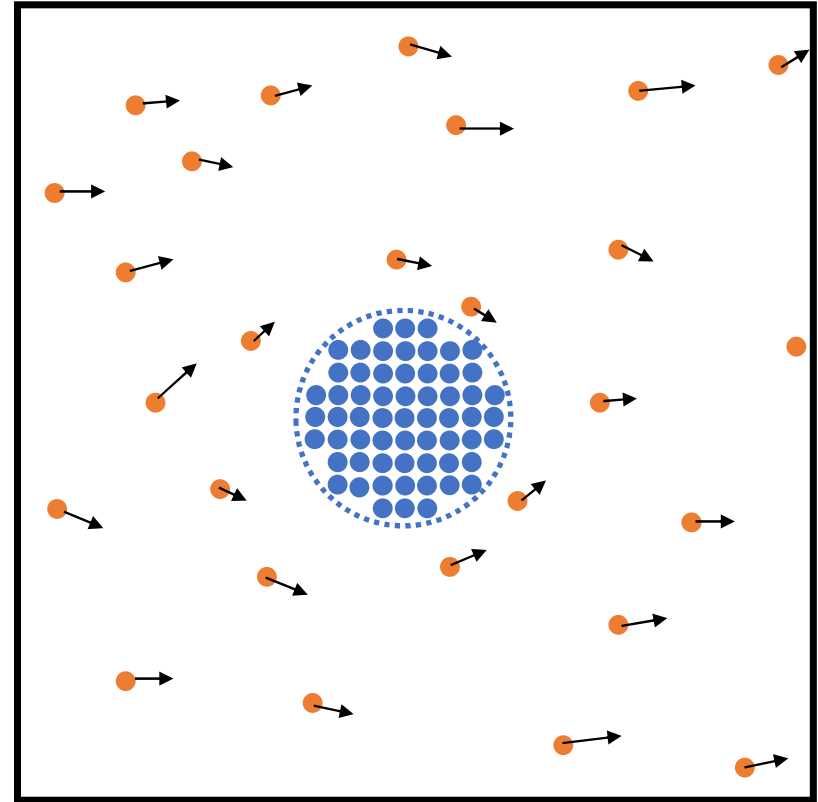
Ghost particles are fixed.

Ghost particles exert forces on **free** particles.

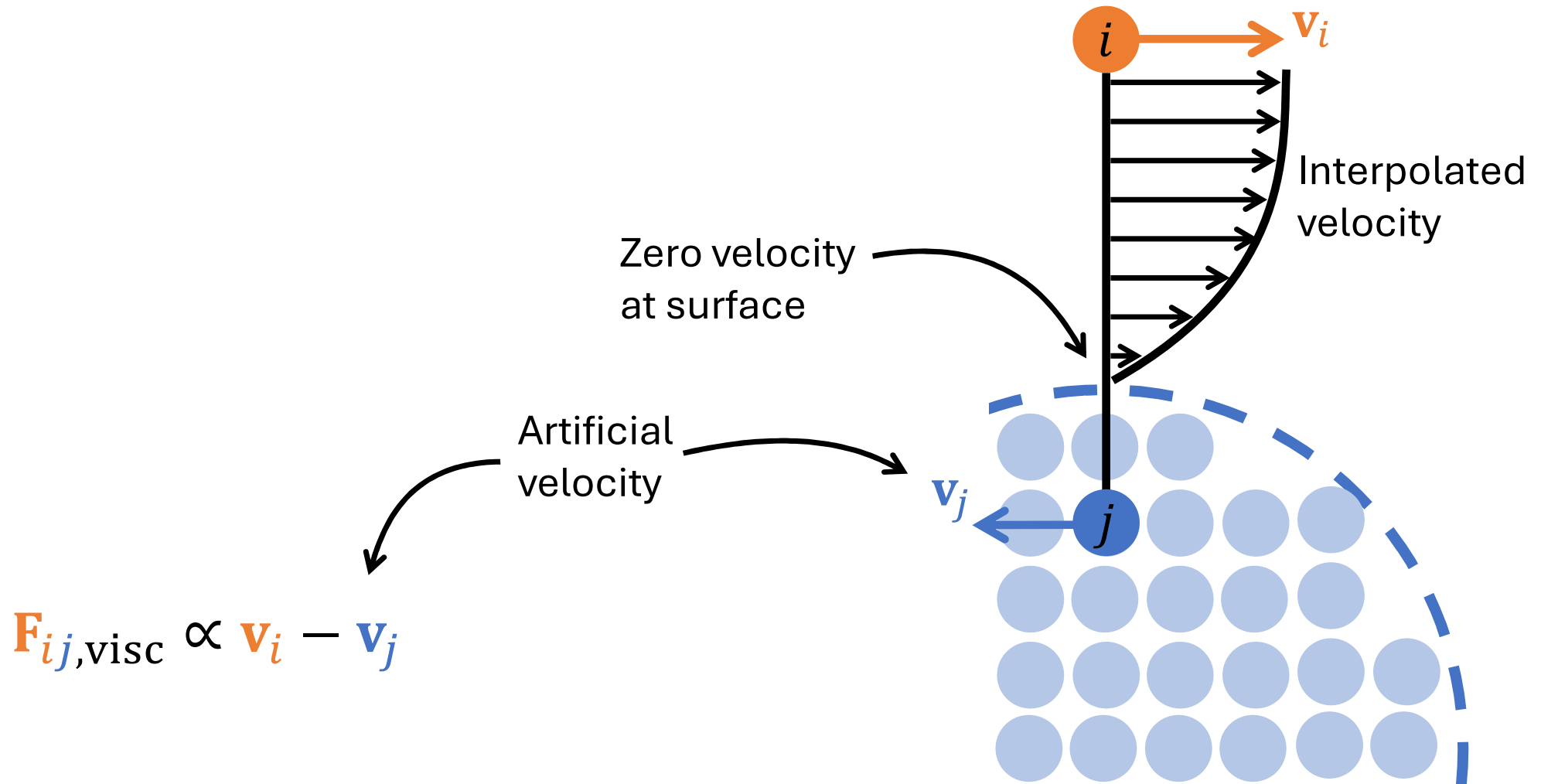
$$F_D = - \sum_{\text{free}} \sum_{\text{ghosts}} F_{\text{ghost on free}}$$

$\searrow$

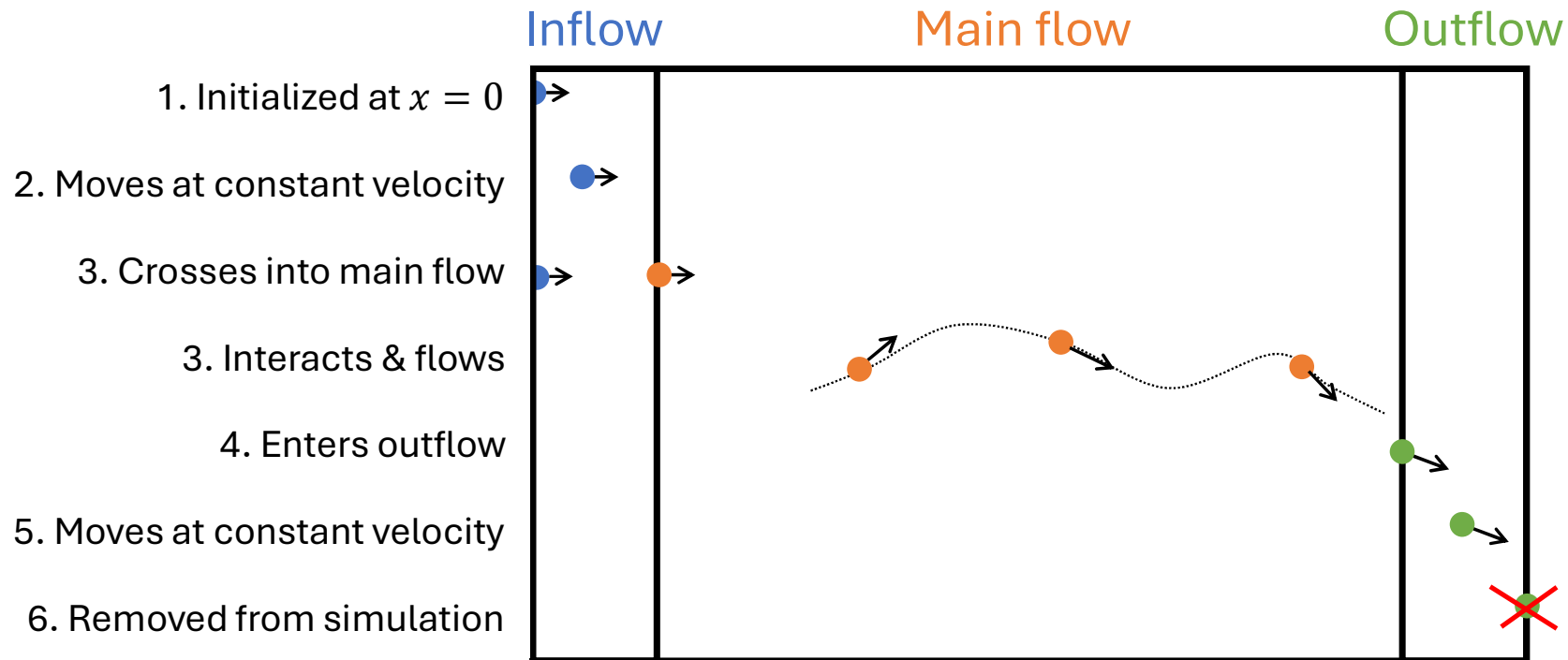
$$c_D = \frac{2F_D}{\rho |\mathbf{v}|^2 A}$$



Artificial velocities enforce the **no-slip boundary condition**

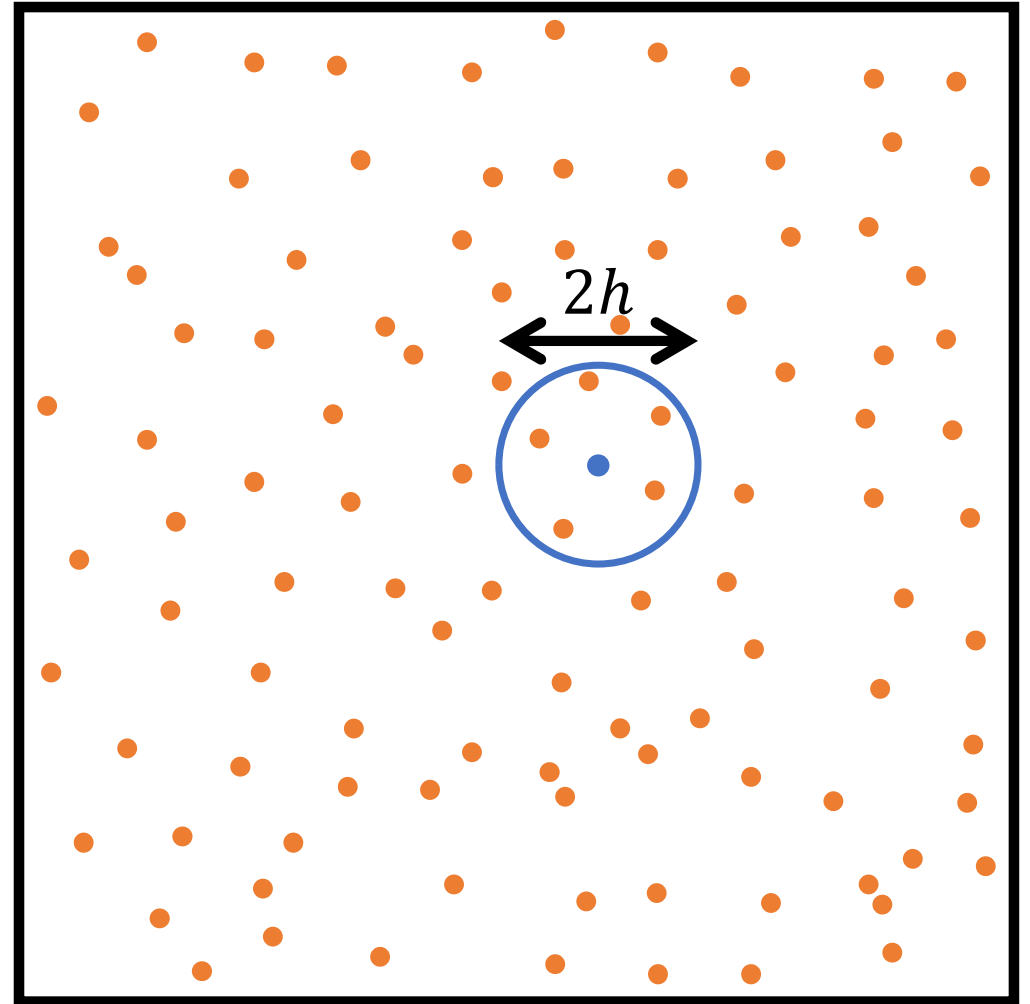


In- and outflow

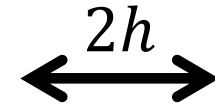


Cell-based algorithm: $O(mn)$ instead of $O(n^2)$

Particles have a finite interaction
radius h

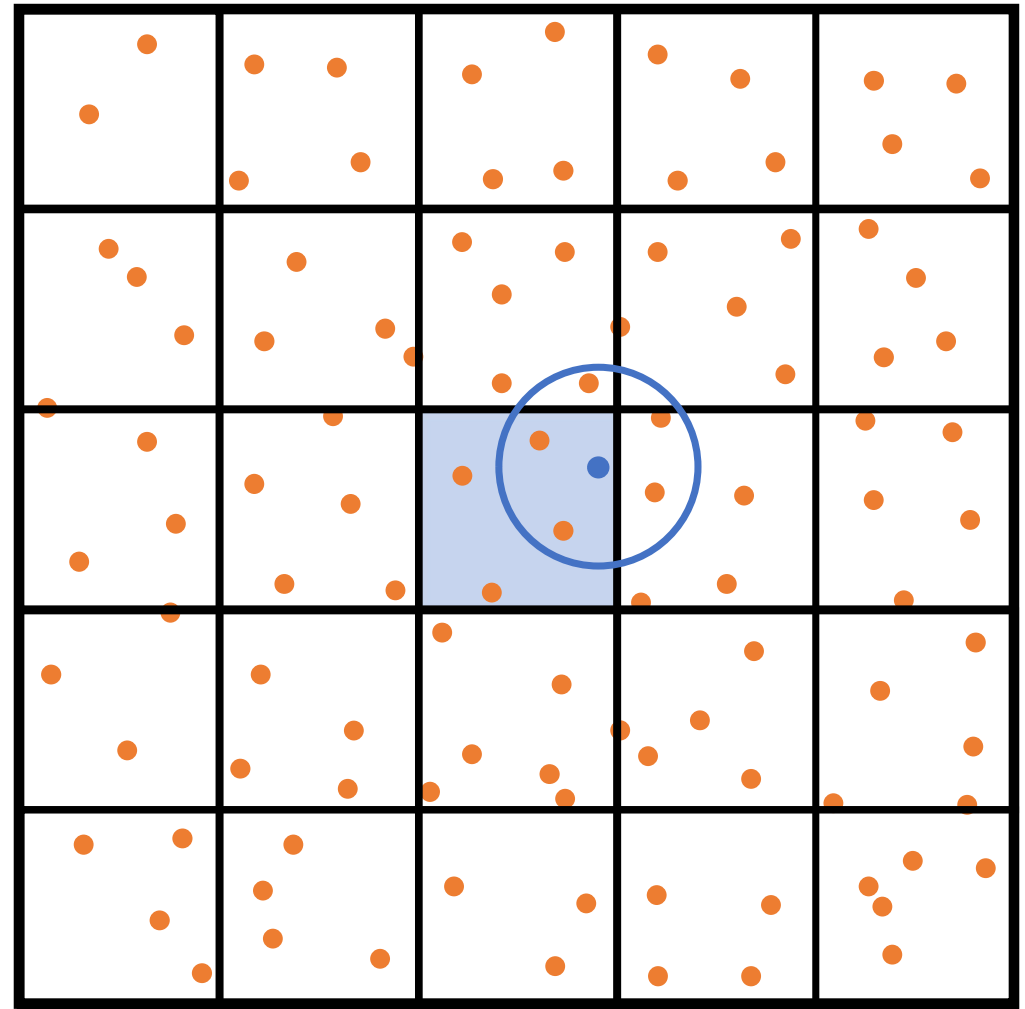


Cell-based algorithm: $O(mn)$ instead of $O(n^2)$

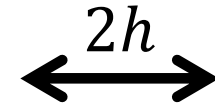


Particles have a finite interaction
radius h

Each particle is in a $2h \times 2h$ **cell**



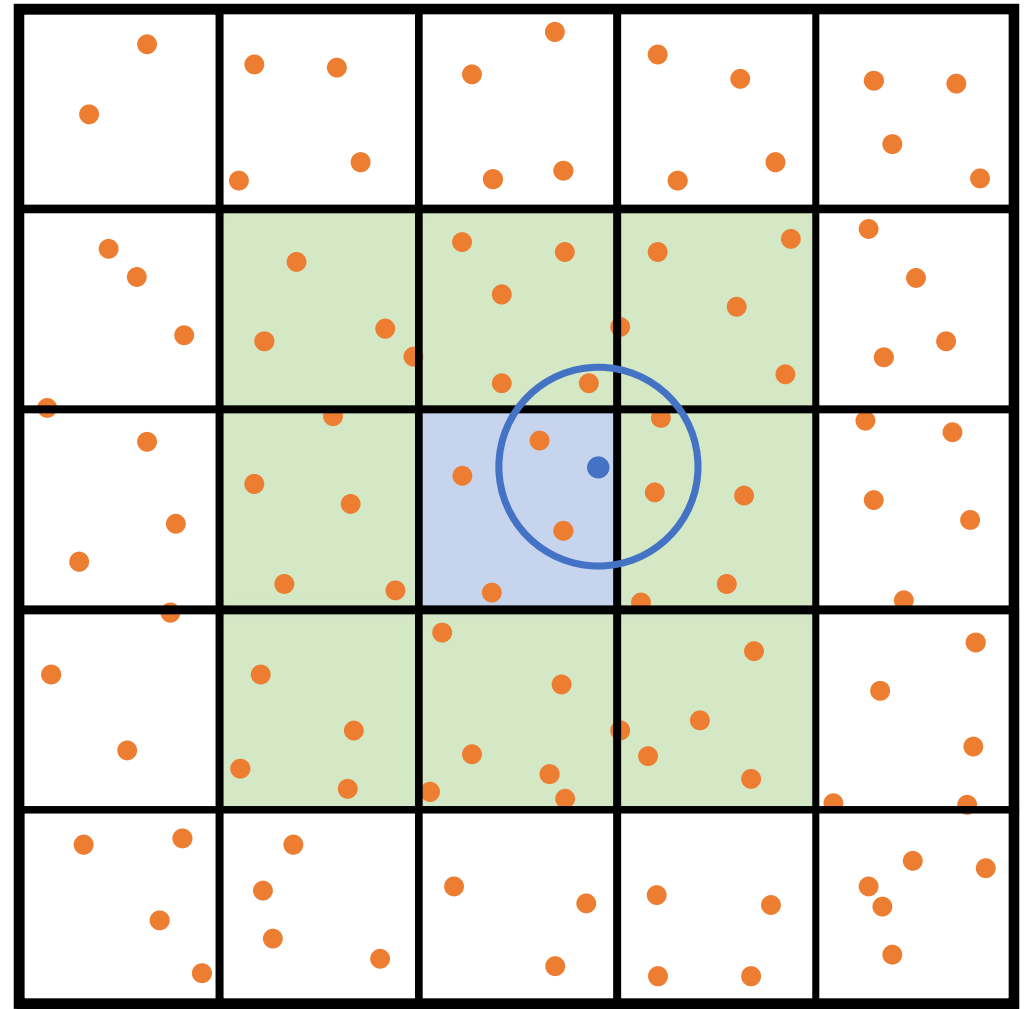
Cell-based algorithm: $O(mn)$ instead of $O(n^2)$



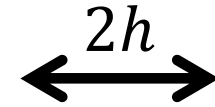
But particles have a finite interaction radius h

Each particle is in a $2h \times 2h$ **cell**

It only interacts with particles in **neighbor cells**



Cell-based algorithm: $O(mn)$ instead of $O(n^2)$



But particles have a finite interaction radius h

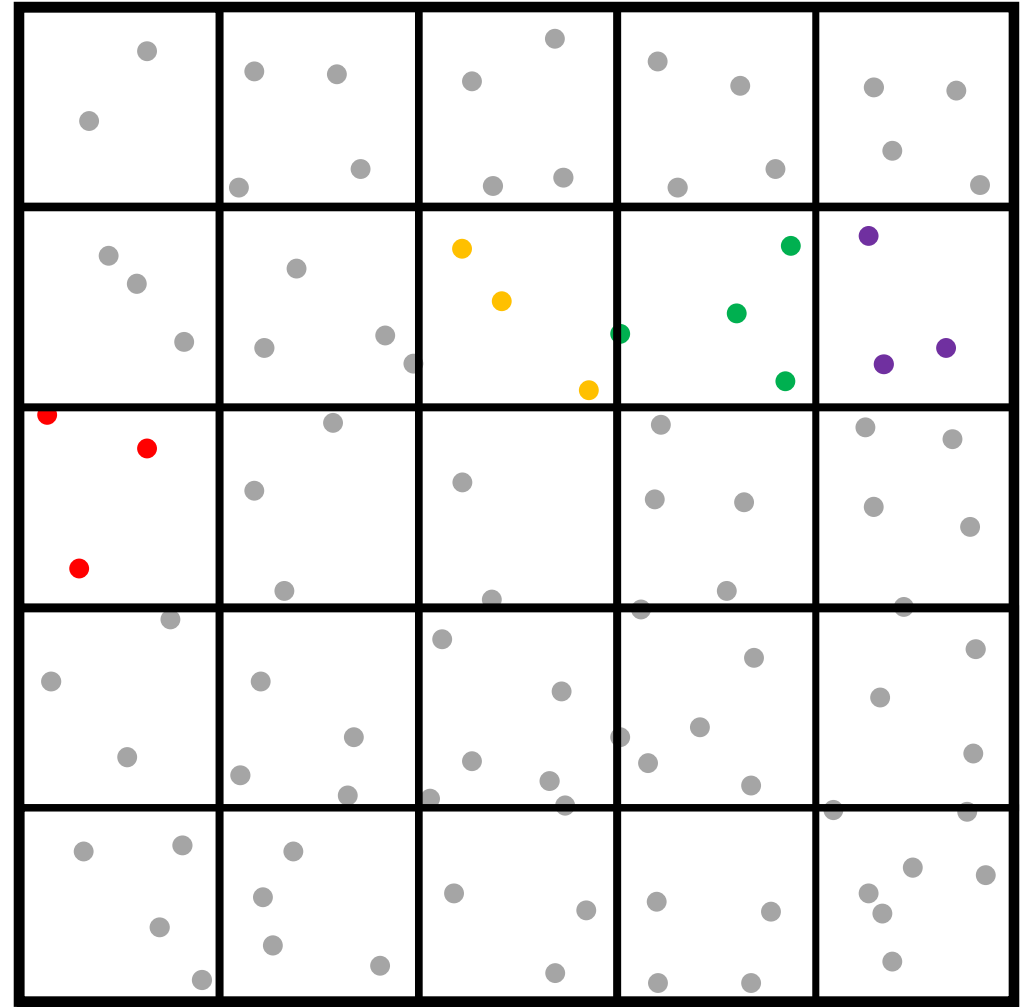
Each particle is in a $2h \times 2h$ **cell**

It only interacts with particles in **neighbor cells**

Particles are stored in a list, sorted by cell:

[..., $p_1, p_2, p_3, p_4, p_5, p_6, p_7, p_8, p_9, p_{10}, p_{11}, p_{12}, p_{13}, \dots]$

This list is re-ordered after each integration step



Courant-Friedrichs-Lewy conditions

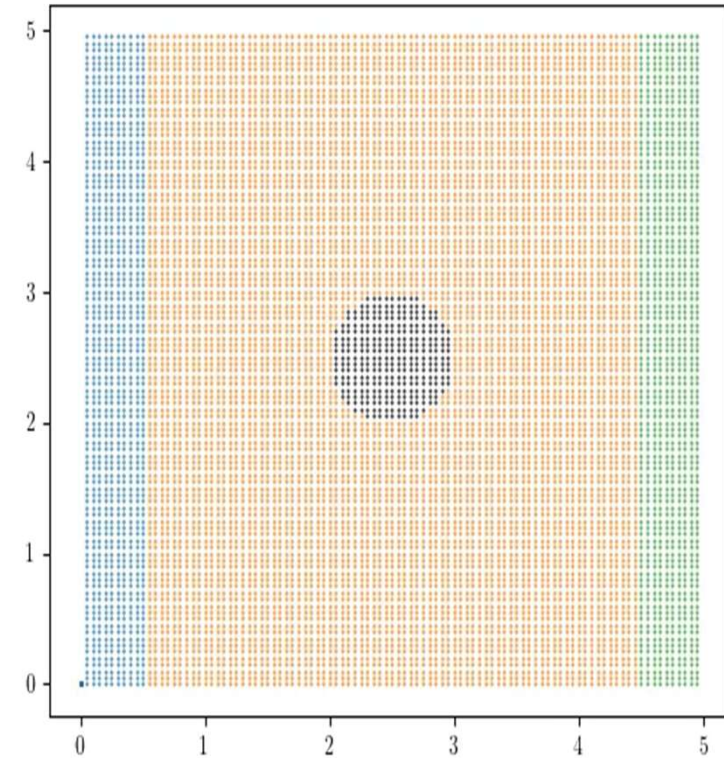
Speed of sound: $\Delta t \leq 0.25 \frac{h}{c_s}$

Viscosity: $\Delta t \leq 0.125 \frac{h^2}{\nu}$

Acceleration: $\Delta t \leq 0.25 \sqrt{\frac{h}{a}}$

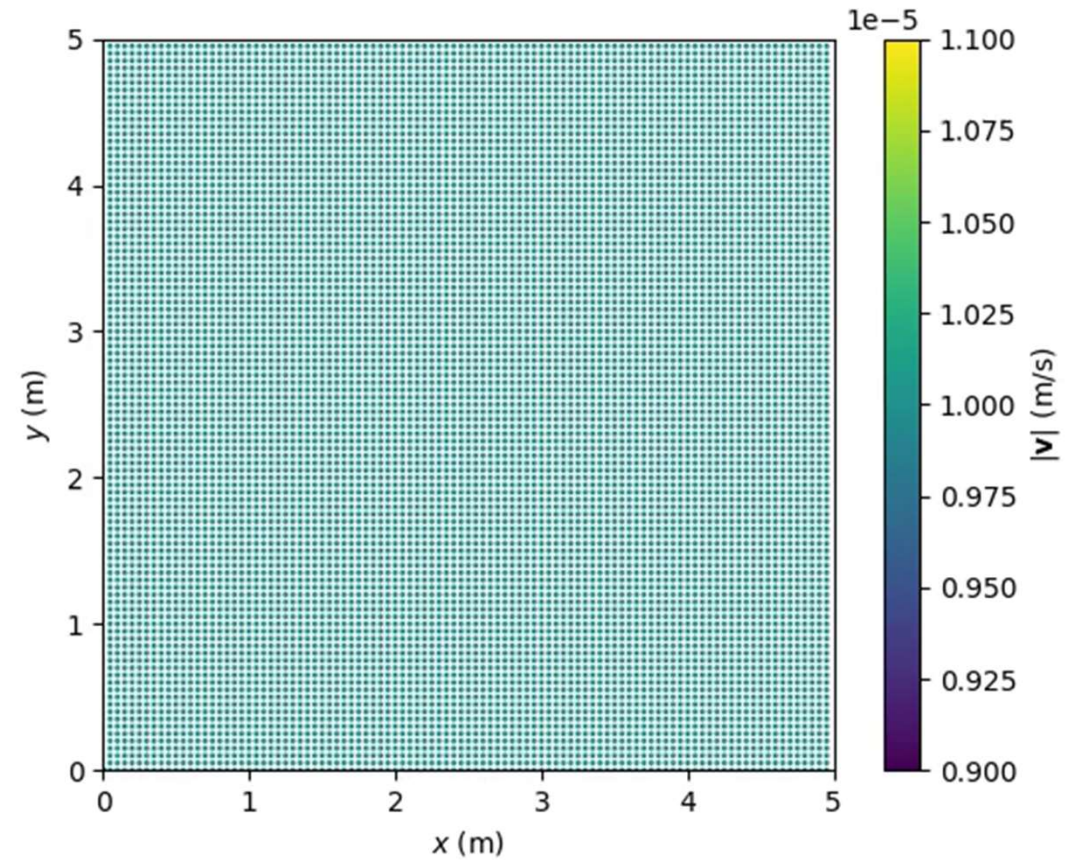
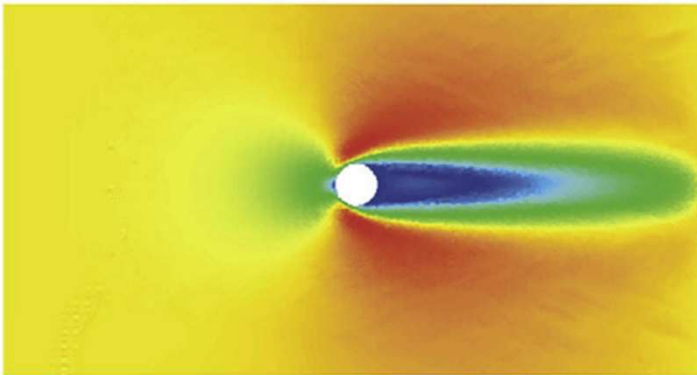
Animated simulation result

- In- and outflow works as expected
- Ghost particles do not move
- Particles flow around & stick to central object



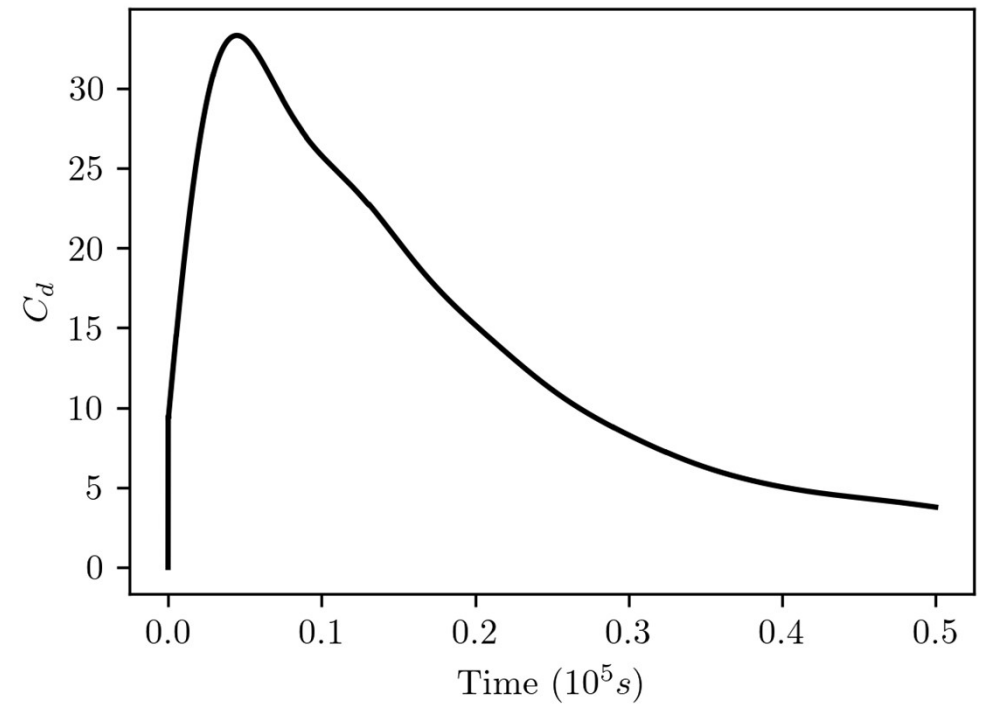
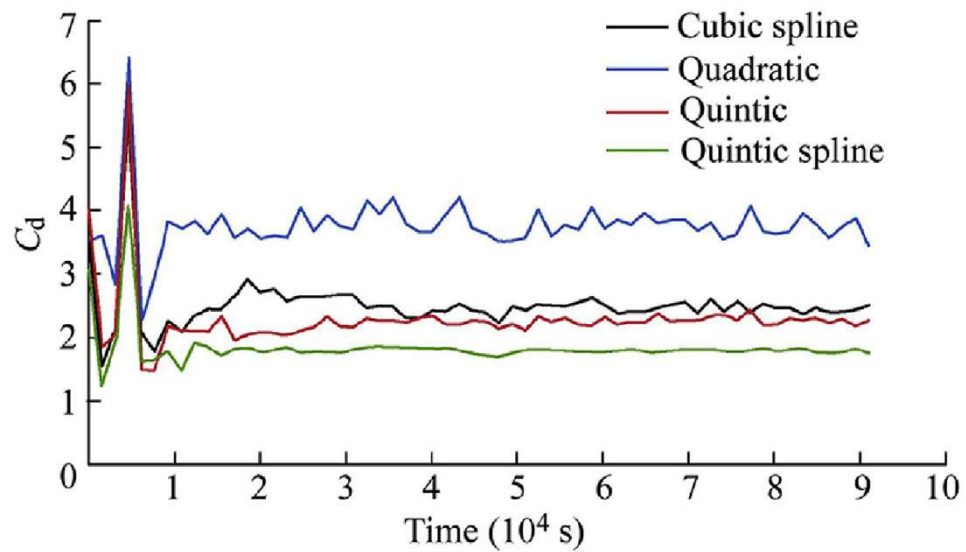
Velocity field

- Particles move as expected
- “stick” to central object of ghost particles



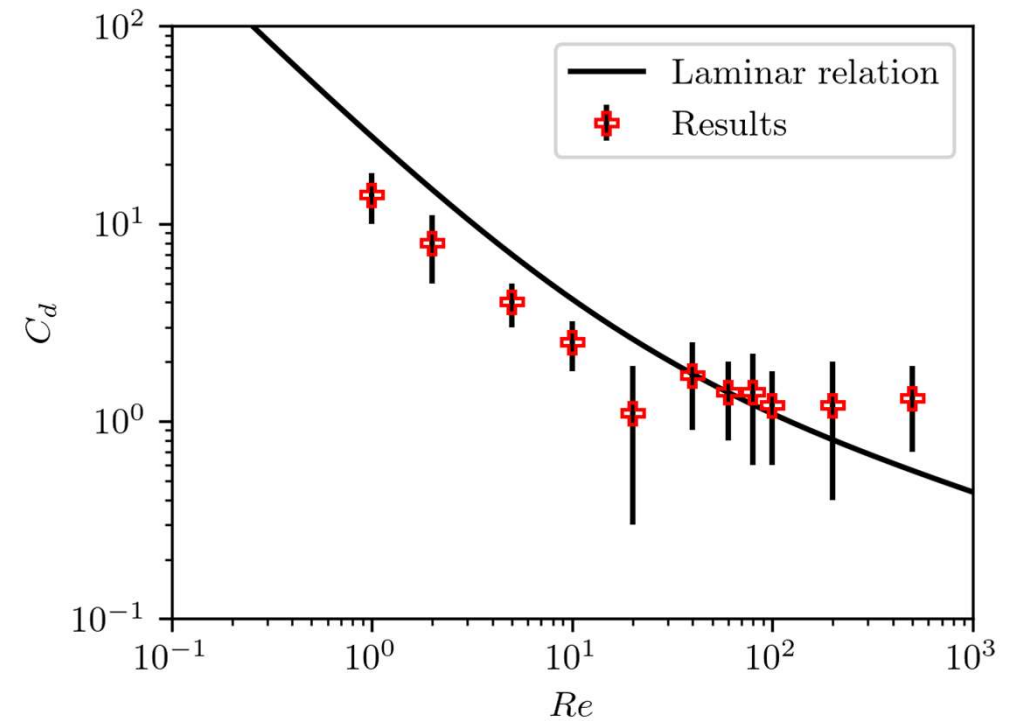
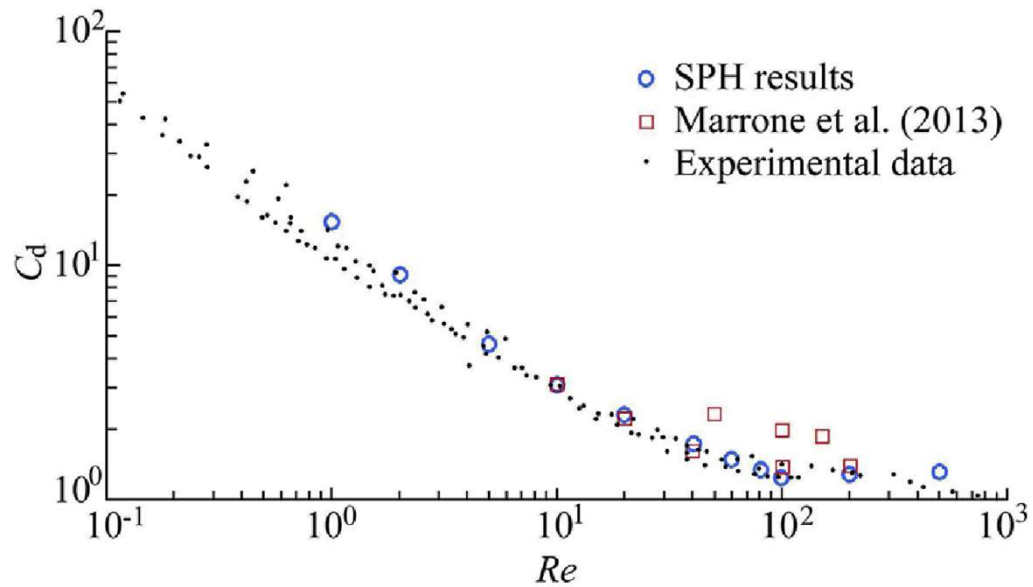
Cd vs. time

- Has a peak in the beginning
- Fluctuates/converges over time



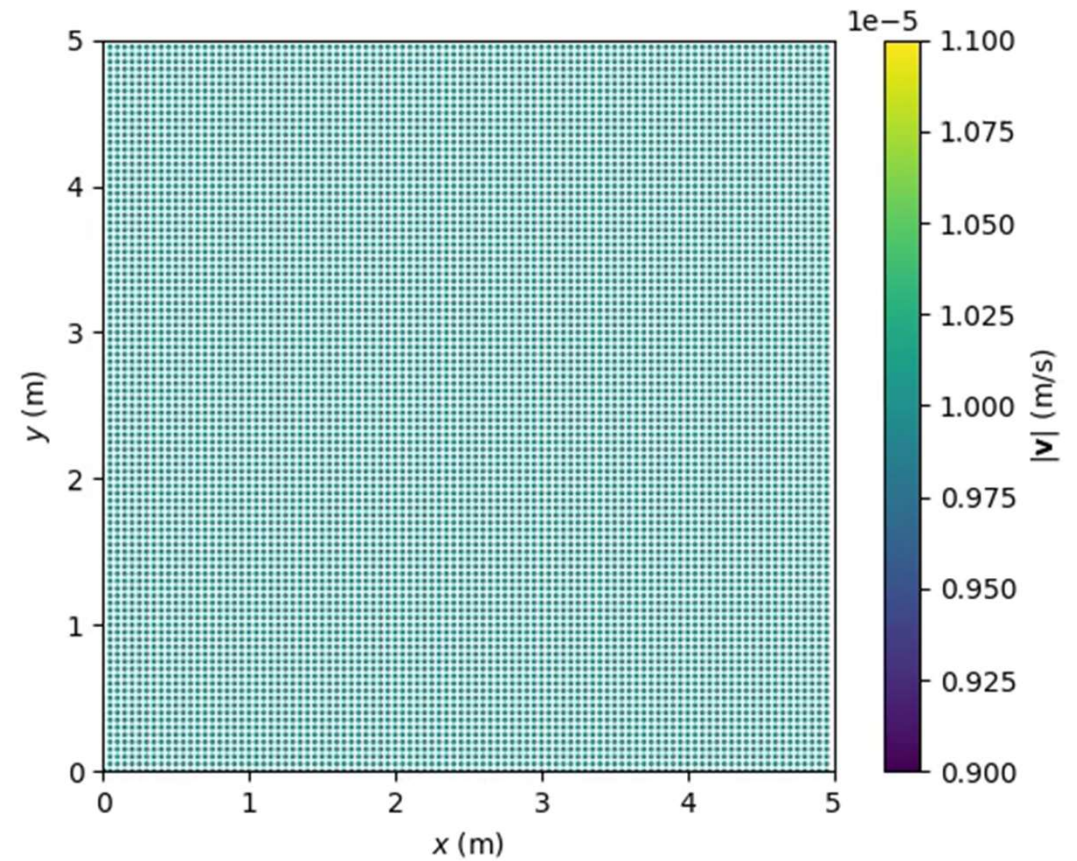
Cd vs. Reynolds number

$$Re = \frac{|\mathbf{v}|L}{\nu}$$



Conclusions

- Works well
- More compute needed
- Hard to get stable



References

Morris, J. P., Fox, P. J., & Zhu, Y. (1997). Modeling low Reynolds number incompressible flows using SPH. *Journal Of Computational Physics*, 136(1), 214–226. <https://doi.org/10.1006/jcph.1997.5776>

Korzani, M. G., Galindo-Torres, S. A., Scheuermann, A., & Williams, D. J. (2017). Parametric study on smoothed particle hydrodynamics for accurate determination of drag coefficient for a circular cylinder. *Water Science and Engineering*, 10(2), 143–153. <https://doi.org/10.1016/j.wse.2017.06.001>

Monaghan, J. J. (2005). Smoothed particle hydrodynamics. *Reports on Progress in Physics*, 68(8), 1703–1759. <https://doi.org/10.1088/0034-4885/68/8/r01>